Design Code

```
`timescale 1ns / 1ps
module traffic light_ controller(
 input clk,
 input reset,
 input pedestrian,
 output reg [2:0] highway,
 output reg [2:0] side _ road,
 output reg ped _ walk
);
//----------------------------------
// Light Encoding
//----------------------------------
parameter RED = 3'b100;
parameter YELLOW = 3'b010;
parameter GREEN = 3'b001;
//----------------------------------
// FSM States
//----------------------------------
parameter HW_GREEN = 2'd0;
parameter HW_YELLOW = 2'd1;
parameter SR_GREEN = 2'd2;
```

```
parameter PED_STATE = 2'd3;
reg [1:0] state;
reg [3:0] timer;
//-------------------------------------
// FSM
//-------------------------------------
always @(posedge clk or posedge reset)
begin
 if(reset)
 begin
 state <= HW_GREEN;
 timer <= 0;
 highway <= GREEN;
 side _ road <= RED;
 ped _ walk <= 0;
 end
 else
 begin
```

```
timer <= timer + 1;
 case(state)
 HW_GREEN:
 begin
 highway <= GREEN;
 side _ road <= RED;
 ped _ walk <= 0;
 if(timer == 5)
 begin
 timer <= 0;
 state <= HW_YELLOW;
 end
 end
 HW_YELLOW:
 begin
 highway <= YELLOW;
 side _ road <= RED;
 ped _ walk <= 0;
```

```
if(timer == 2)
 begin
 timer <= 0;
 if(pedestrian)
 state <= PED_STATE;
 else
 state <= SR_GREEN;
 end
 end
 SR_GREEN:
 begin
 highway <= RED;
 side_road <= GREEN;
 ped_walk <= 0;
 if(timer == 5)
 begin
 timer <= 0;
 state <= HW_GREEN;
 end
 end
```

```
PED_STATE:
 begin
 highway <= RED;
 side_road <= RED;
 ped_walk <= 1;
 if(timer == 4)
 begin
 timer <= 0;
 ped_walk <= 0;
 state <= SR_GREEN;
 end
 end
 endcase
 end
end
endmodule
```